

Frogs and Toads Puzzle

Below is the $n = 3$ version of the puzzle.



Let m_n be the minimum number of moves (jumps or slides) in a winning strategy. Can you express m_n as a recurrence relation? What's its generating function? Can you find a closed form solution?

Non-Homogenous Linear

What are the recurrence relations, generating functions, and closed form solutions for c_n , T_n , and a_n ?

Sequence 1: $1, 1, 1, 1, \dots$, $b_n = b_{n-1}$, $b_0 = 1$

Sequence 2: $1, 2, 3, 4, \dots$, $c_n = \sum_{i=0}^n b_i$

Sequence 3: $1, 3, 6, 10, \dots$, $T_n = \sum_{i=0}^n c_i$

Sequence 4: $1, 4, 10, 20, \dots$, $a_n = \sum_{i=0}^n T_i$

$\vdots$

Sequence n: ???

What is sequence n? recurrence? generating function? closed form?

From the homework/class

Ternary strings with double zeros.

or

Runs of 1s in binary strings